

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 08- 102- 2

Date: 08-08-2023

EDMS 08-102-2

TECHNICAL SPECIFICATION

FOR

EPOXY SOLID-CAST/VPE OR CAST COIL

DRY - TYPE DISTRIBUTION TRANSFORMERS

10.5-11/0.4 KV, 50 KVA up to 5000 KVA

Issue: Aug. 2023 / Rev- 2

❖ توجد بنود اختيارية (Optional items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

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1. SCOPE:

This specification covers the minimum technical requirements for design, engineering, manufacturing, packing, installation, and testing of an epoxy solid-cast resin/VPE coil (10.5-11/0.4KV) dry-type transformers (indoor installation) for 50 HZ rated frequency network with enough detail to ensure that true solid-cast/VPE coil (solid cast accepted) transformers will be offered.

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
IEC 60076 Power Transformers	Part 1 General
	Part 2 Temperature rise for liquid-immersed transformers
	Part 3 Insulation levels and dielectric tests
	Part 4 Lightning impulse and switching impulse testing
	Part 5 Ability to withstand short circuit
	Part 10 Determination of sound levels
	Part 11 Dry type power transformers
	Part 12 Loading guide for dry type power transformers
	Part 20 Energy Efficiency
IEC 60214 Tap-changers	Part 1 Performance requirements and test methods
	Part 2 Application guidelines
ISO 9001	International standard for a quality management system (QMS)

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3. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required transformers shall be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +45°C (50°C as option according to ...EDC) (35°C monthly average of the hottest month, 25°C yearly average)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above sea-level

4. DESIGN CRITERIA

4.1 Rated Power:

- The transformer shall be able to deliver the maximum continuous power rating for an unlimited time at the main tapping of the tap changer without exceeding the allowable temperature rise limit, if the applied voltage is equal to the rated voltage and maintained at its rated frequency.
- The maximum continuous power rating is selectable from the set of standard rates given in Table (5).

4.2 Voltage Transformation Ratio

- The transformation ratio shall be (10500/400 V) or (11000/400 V) according to EDC requirement. The deviation of the transformation ratio measured at no-load and 50 Hz should not exceed 0.5% from its nominal value.
- The transformer shall be able to work properly without damage as long as the voltage-to-frequency ratio does not exceed the corresponding ratio at rated voltage and rated frequency by more than 5%.

4.3 Vector Group

- The winding connection group of the transformers shall be delta on the primary side and star on the secondary side in accordance with the vector group Dyn11 as specified in IEC 60076.

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4.4 Tapping Range

- a) The taps are rigidly supported by cast into the epoxy along the central face of each high voltage coil; taps are changed by moving the flexible bolted links from one connecting point to the other, and tap terminals are clearly numbered and referenced on the transformer nameplate.
- b) Seven positions Tap-changer and the associated operating mechanism shall be able to vary the transformation-ratio in steps of 2.5%. The range of transformation ratio shall be +5% to -10%.
- c) The tap-changer enclosure shall be equipped with a suitable transparent window of thickness not less than 3 mm to grant observation of the current tapping.
- d) Voltage tapping shall be provided on the high voltage side of the transformer. Direction of operation for Raise/Lower tap shall be provided.

4.5 On-Load Tap Changer (Optional as per ... EDC Requirements)

- a) In case of required on-load tap-changer (OLTC), the operating mechanism shall be equipped with three-phase motor of 400 V and 50 Hz. All auxiliary supply voltage shall be DC 110 V.
- b) The voltage sensing equipment should prevent operation of the tap-changer if the reference voltage is lost.
- c) The relays, switches and push buttons shall be enclosed in a weatherproof cubicle and be readily accessible for maintenance at ground level with the transformer energized.
- d) The Motor Drive Unit of the OLTC should contain a thermostatically controlled anti-condensation heater, a door operated light and a manual handle for emergency and maintenance operation.
- e) It shall be weatherproof and dust proof (IP54) and shall be bolted to the transformer main tank in a convenient position such that the operator standing at ground level can carry out maintenance on all equipment contained in the cabinet with the transformer energized.
- f) Tap changer mechanism should send signal to initiate Indication/Alarm for the following:
 1. Tap changer in progress
 2. Tap changer incomplete
 3. Out of step Indication
 4. Supply voltage failure
 5. Tap position
 6. Tap changer counter

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4.6 Overload Capacity

- Permissible overload capacity of air natural (AN) partial cast dry-type transformer shall be in accordance to IEC 60076-12. For transformer with thermal class (F), the permissible overload capacity shall be as listed in Table (3).

Table (3)

Previous continuous loading (% of rated power)	Windings temperature/ hot spot	Duration of overloading for specific levels of overloading (% of rated power), Maximum temperature for hot spot 145 °C				
		10%	20%	30%	40%	50%
50 %	46 °C /54 °C	50 min	45 min	30 min	25 min	20 min
75 %	79 °C /95 °C	35 min	25 min	17 min	13 min	10 min
90 %	103 °C /124 °C	15 min	8 min	6 min	5 min	3 min

4.7 Short-circuit Withstand

The transformer shall be capable to withstand, without damage, the thermal and mechanical effects of short-circuit levels for the duration given in Table (4).

4.8 Temperature Rise Limits

The transformer shall be able to deliver its maximum continuous ratings without exceeding the following permissible temperature rise limits given in Table (4).

4.9 Insulation Level

The test voltages given in Table (4) shall be applicable to the transformer.

4.10 Impedance Voltage

- The short-circuit impedance (or impedance voltage Z% expressed as a percentage of rated voltage) measured at mid-tap at 75 °C shall be as follows:
 - 4 % for transformers of power rates (from 50 kVA up to 630 kVA)
 - 5 % for transformers of power rates (800 – 1000 – 1250 kVA)
 - 6 % for transformers of power rates (1500 – 1600 – 2000 – 2500 kVA)
 - 7 % for transformers of power rates (3150 – 5000 kVA)

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- b. According to IEC 60076-1, the measured short-circuit impedance is subjected to a tolerance of $\pm 10\%$ of the specified value.

Table (4)

Transformer kVA Rating	As given in Table (5) (1 p.u for AN / 1.4 p.u for AF)
Cooling Class	AN / AF
Type	Ventilated dry-type
Phases	3
Frequency	50 Hz
Taps	Off Load $+2 \times 2.5\%$, $-4 \times 2.5\%$, (OLTC is optional) ^(*)
Average winding temperature rise above ambient temperature at rated current "C"	95°C for 45°C ambient or 90°C for 50°C ambient (120°C is optional) ^(*)
Insulation system temperature HV&LV coil	155°C (class F) / 220°C ^(*)
Ambient temperature	45°C, (50°C is optional) ^(*)
Primary Voltage	10.5 kV or 11 kV
Primary Connection	Delta
Secondary Voltage (at No load)	0.4 kV
Secondary connection	Star
Vector Group	Dyn11
Insulation level	Primary rated lightning impulse withstand voltage (kV _{peak})
	75 kV
Duration of short circuit	Rated power frequency withstand voltage (kV _{rms})
	28 / 5 kV
Duration of short circuit	Thermal (rms)
	2 seconds
Duration of short circuit	Dynamic (peak)
	0.5 s (Cat. 1), 0.25 s (Cat. 2) $\pm 10\%$
S.C. level of 10.5 / 11 kV network	500 MVA
Impedance voltage at (75°C) at main tap	As given in Table (5)
Noise level	As given in Table (5)
Partial discharge	≤ 10 pC
Climate class	C1
Environmental class	E2
Fire protection	F1

^(*) Option items as per ----EDC requirement.

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4.11 Primary Windings

- a. The high voltage coils shall be wound with high conductivity aluminum conductors (Copper is optional as per EDC requirements) and then Solid-cast (Encapsulated under Vacuum and pressure) separately in molds.
- b. Coils shall be separated by uniform air space that permits free circulation of cooling air between the coils.
- c. Windings shall be properly sized, insulated and supported for the line and load voltages and currents to be encountered in operation including short circuits, when the transformer is properly installed using 180° (class H), rated insulated material and adequately protected.
- d. Construction of the transformer should separate the primary connections and the secondary connections by placing them on opposite sides of the core.
- e. The transformer core limb shall be grounded to the base through a copper strap, and taps shall be provided on the primary.

4.11.1. Encapsulation (For primary coil)

- The Encapsulation process should involve pre-heating the coils and then filling each mold with degassed and thoroughly mixed epoxy under vacuum.
- Once filled, the coils should then be oven cured to insure a solid void free casting.
- The oven curing process shall be controlled and monitored to ensure that each coil is properly cured.
- The high voltage coils should withstand the half and full wave impulse tests as well as partial discharge tests to insure a properly cured and void-free casting.
- The epoxy should contain filler material providing characteristics superior to unfilled epoxy including higher temperature rating, better heat conductivity, better arc resistance and adhesion to the conductor, plus a coefficient of expansion closer to that of the conductor material.

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4.12 Secondary Windings

1. The LV windings will be of non-encapsulated design with multi-layer full sheets of Aluminum foil or copper foil in order to cancel out axial stress during short circuit.
2. This foil shall be wound together with an interlayer insulation of pre-impregnated resin and thermally cured in an oven to achieve its thermal, mechanical and moisture penetration properties as per IEC 60076-11 requirements that are comparable, for LV windings, with those of cast windings.
3. Foil will be protected everywhere with a class F or class H insulation material even in the air ducts.
4. It shall be to seal both edges (top and bottom) of the windings, that will prevent the entry of dust or moisture inside the coil.
5. LV Winding shall be designed and adequately insulated and supported to withstand the short circuit forces.
6. The low coils should be wound with high conductivity aluminum conductors (Copper is optional as per EDC requirements).
7. The fabricated process for the insulation should be vacuum pressure encapsulated (VPE) or the fabrication process for the insulation (cast).
8. Coils should be separated by uniform air space that permits free circulation of cooling air between the coils.

4.12.1 Impregnation process (in case of VPE):

A. Pre-heat:

The windings are pre-heated in an oven or chamber to initiate the removal of moisture from the insulation and to aid in the subsequent impregnation of the resin. This is accomplished since the resin viscosity is lowered as the resin makes contact with the heated material.

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B. Dry vacuum cycle:

During the dry vacuum cycle, gases and moisture are removed from the insulation materials and winding.

C. Vacuum immersion (resin-encapsulated):

While under vacuum, the resin is introduced as a liquid state into the processing tank (vacuum immersion), until the winding is completely submerged.

D. Vacuum hold cycle:

The vacuum is maintained for a period of time to encourage the resin to impregnate the insulation and to remove air from the resin.

E. Pressure cycle:

Shortly following the vacuum hold cycle, the processing tank is released, then pressurized on the free surface of the liquid bath to force the resin to impregnate the insulation voids.

F. Curing:

- The winding or the tank is removed, and the winding is allowed to drip (dry) prior to being placed in an oven where the resin is cured to a solid.
- The construction of low voltage windings should be modular design for repairing and maintenance during operation.
- Construction of the transformer should separate the primary connections and the secondary connections by placing them on opposite sides of the core.
- The transformer core limb should be grounded to the base through a copper strap, and taps should be provided on the primary.

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4.13 Core

- The core shall be constructed of the best quality, low loss; high permeability cold rolled grain-oriented steel laminations, on-aging silicon steel insulated on both sides.
- It shall be miter-corner-cut without burs and hand stacked into a cruciform arrangement of different width of laminations, to provide a cross-section most nearly assembling the circular cross-section inside the low voltage coil.
- Laminations shall be cut and stacked utilizing step-lap construction to reduce losses and sound.
- Designed flux density shall be kept well below saturation (≤ 1.7 Tesla).
- The assembled core shall be braced with heavy structural steel angle or channel to apply uniform clamping forces across the entire width, top and bottom.
- The core should have steel straps running top to bottom down the front and back of each core leg, inside the coils.
- Horizontal and vertical bracing shall be through bolted to insure consistent clamping forces on all parts of the magnetic steel laminations.
- The core shall be protected from corrosion with a higher temperature rated rust-resistant coating.
- The temperature of the core, metallic parts and adjacent materials should not reach a value that will cause damage to any part of the transformer.
- The core assembly should enable the removal of the coils in the field, if this shall become necessary.
- The entire core assembly shall be covered with a resin-based lacquer for corrosion protection before the coils are mounted.

4.14 Core and Coil Assembly

- Construction should consist of separate high and low voltage coil for each phase, mounted coaxially with high/low air space between coils adequate for the rated voltage potentials.
- Coils shall be mounted concentrically and supported in place on the transformer core to restrain movement in shipping and in services.

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- Each coil shall be supported on epoxy or equivalent fiber glass blocks withstand mechanical and thermal strength, top and bottom, with resilient pads to retain the coils while permitting thermal expansion under loading.
- The core and coil assembly shall be designed and manufactured to withstand without damage the short circuit testing as defined in IEC standards.

4.15 Busbar Connection

- Lead connection of aluminum bar is provided by cold-press technology.
- All bus work shall be designed to withstand maximum short circuit stresses when connected to a supply system having a fault capacity of 25 KA for 3 sec., symmetrical at rated voltage, and to meet all test voltage requirements, without audible or visible corona.
- Busbar shall be sized to maintain a current density of less than or equal 1.5 amps per mm² at 100% name plate load and provided with connector plates bimetal AL/CU.
 - a. Low voltage busbars connection shall be high conductivity material, Connections of coil windings to busbar shall be brazed or welded using shielded arc techniques or friction welding technique designed to provide high joint integrity. Connection and terminal bolting surfaces shall be tin or silver-plated. Bolted connections should include spring washers to maintain bolted surface pressure. All busbars shall be sized to conform to industry standards and practices, for the amperes involved and maximum specified ambient temperatures.
 - b. High voltage busbar shall be high conductivity material, and fixed with three epoxy resin HV insulator with 2 cm/KV min. Leakage path with bolt-connected joint and terminal surfaces tin or silver plated.
 - c. High voltage leads delta connection shall be busbar or flexible cable insulated with technically accepted types of terminations. The tap's leads shall be flexible cable or solid links or jumpers (as per ---EDC requirement).
 - d. Ground busbar shall be copper with 150 mm² C.S.A, extending full length of two opposite sides of the enclosure. It should have provisions to mount appropriate terminal connection for adjacent switchgear or external grounding.

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- e. A galvanized shielded copper ground shall be applied to facilitate the grounding of each metal part with the following specs:
1. Copper: 99.9 % purity minimum.
 2. Width and Thickness: 3 cm *1.5 mm at least. (effective C.S.A at least 35mm²)
 3. Flexibility: full.
 4. Number of Shielded conductors to ground the transformer body: 2 .

4.16 Enclosure (Optional as per ... EDC Requirements)

- a. Enclosure's dimensions shall be suitable for installation in transformer's room with IP23.
- b. Transformer with IP00 shall be mounted in steel kiosk with IP23 to knockdown enclosure with anti-corrosion treatment (electrostatic painting).
- c. Enclosure should have a removable front face to gain access to the high voltage connection equipment with two easy handles.
- d. Entrance Cables: from up or/and down of the enclosure by reinforced aluminum material of thickness not less than 3 mm.
- e. Enclosure thickness for all sides, cover and bottom shall be not less than 2.5 mm before electrostatic painting.
- f. Enclosures shall be suitable for complete transformer lifting with Eye Hooks or Lifting Eyes.
- g. The factories should make the design and beams to approve that the total weight of the unit can be lifted from enclosure without removing the whole ceiling of it.
- h. The enclosure shall be equipped with (3) suitable transparent windows of thickness not less than 3 mm (one for each phase), and shall be against L.V. side of the transformer. (door is accepted with maintained IP23)
- i. The enclosure shall be provided with two adequate fans to ensure uniform heat transfer (Option as per __EDC requirements).

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4.17 Base

- a. The transformer base shall be a separate rigid steel weldment capable of supporting the core-and-coil assembly with enclosure.
- b. Construction should include provisions for lifting as well as for jacking and skidding or rolling in both directions.
- c. Two stainless steel ground pads (nuts) with two tapped holes for attaching ground leads shall be provided, one in each diagonally opposite corner of the base.
- d. Lifting provisions shall be designed for lifting the transformer and the core-and-coil assembly, separately or together.

5. ACCESSORIES

- a. It is not necessary to approve the transformer's accessories separately. The approval issued for the transformer with the inclusions is sufficient.
- b. An instantaneous and maximum temperature – indicating thermometer shall be provided on the front of transformer.
- c. The transformer and their accessories shall be able to withstand the effect of the rated short circuit through windings for 2 seconds duration.
- d. The transformer and their accessories shall be able to withstand the effect of the dynamic short circuit current through the windings for (0.5 s (Cat. 1) from 25 kVA up to 2500 kVA, 0.25 s (Cat. 2) from 2501 kVA up to 100 MVA) $\pm 10\%$.
- e. The transformer should have a weatherproof nameplate of stainless-steel material placed in a clear position.

5.1 Thermal protection on LV windings and/or core

- a. Transformer shall be provided with six fans to give a forced air-cooled rating above the self-cooled full rating, miniature circuit breaker and control wiring (wire markers included).
- b. A thermo-resistance PT100 (Platinum "100" for the resistance in ohms at 0 °C) shall be connected to a suitable electronic device.
- c. PT100 thermo-resistance should allow logging of windings and core and a remote transmission of

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data.

- d. A thermal sensor and a fan controller shall be supplied, installed inside the enclosure.
- e. The temperature controller should log the signal coming from three (3) PT100 sensors at least to indicate the measured temperature.
- f. It should include two relays with adjustable threshold for the alarm and release signals as well as an auxiliary contact to control a possible set of fans for air forced ventilation.
- g. It shall be a microcontroller/microprocessor-based unit specifically designed for dry type and cast resin transformers.
- h. It shall be developed with layout and advantages of the following technology platform:
 - 1. Dual display.
 - 2. Voting function.
 - 3. The controller with increased operational capacity.
 - 4. Data management.
 - 5. HFN (Hourly Fan No cycling)
- i. It should provide in a single product with the following outputs:
 - 1. Analog 4 – 20 mA
- j. Digital RS485 Modbus RTU.
- k. It shall be able to monitor the temperature of the transformer by a remote communication system.
- l. It should in, addition to transmitting real-time temperatures, offer the possibility to vary all the settings including the relays intervention thresholds.
- m. It shall be equipped with three (3) PT100 inputs (at least) to monitor the temperature of the core and windings while sensor inputs shall be available.
- n. It should have the following:
 - 1. ALARM relays to give a signal for high temperatures.
 - 2. TRIP relay to give signal to disconnect the transformer in case it reaches the maximum threshold through C.B, RMU, ... etc.
 - 3. FANS relays to start the ventilation system.
 - 4. FAULT relay to indicate the damage or break of PT100 sensors and the critical failures (trip operation failed).
 - 5. SIGNAL relay for fans failure at BMS (Building Management System)
- o. It shall be supplied with a coating on the electronic cards, resistant to difficult weather conditions,

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particularly characterized by high temperature and humidity.

- p. It shall be universal power supply with input from 24 to 240 Vac/dc.
- q. "The fan controller box must be tightly closed to protect it from dust and impurities" such that the screen and the selector can be accessed from outside without need to open the enclosure.
- r. The thermal sensor and a fan controller shall be used to indicate winding temperature and provide a single-phase voltage to operate fans or to disconnect fans according to the temperature of winding and auxiliary contacts to provide alarm and to trip the M.V.C.B.
- s. All control wiring and thermal sensor must be in ducts and thermal flexible plastic pipes.
- t. The manufacturer must provide a brief user manual (monitoring steps) for the fan-controller and to be installed beside it.
- u. The transformer shall be provided with indicating lamps on the transformer outer surface with a lamp for each fan to indicate whether the fans are working properly or not, with an arrangement of the lamps indicating their dependency for each fan.

5.2 Anti-vibrating pads

- a. Anti-vibrating pads shall be composed of special rubber supports supplied loose to be positioned by the ... EDC under the transformer's wheels. (Noise Reduction Dumper)
- b. They should allow a great reduction of the vibrations transmitted to the structure and therefore of the noise and possibility of structural resonances.
- c. Anti-vibrating supports shall be designed and supplied for special application in compliance with ...EDC specification.

5.3 Ethernet communication channels (Optional)

- a. The Ethernet communication channel shall be a microcontroller/microprocessor-based device.
- b. The Ethernet communication channel shall be designed for dry type cast resin transformers with integrated Ethernet port.
- c. The Ethernet communication channel shall be equipped with all the necessary functions needed to monitor and control the temperature of the transformer.
- d. The Ethernet communication with the network shall be via Modbus TCP/IP protocol allowing to

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display and to program all the unit functions from the comfort of the desk.

- e. The Ethernet communication channel should maintain the traditional PT100 inputs and relays for ALARM, FAULT, TRIP, and FANs.
- f. The unit shall be supplied with a coating on the electronic cards, resistant to difficult weather conditions, particularly characterized by high temperature and humidity.
- g. The protection relays shall be universal power supply with input from 24 to 240 Vac / Vdc.

5.4 Ventilation system (set of electric fans for air forced (AF) cooling)

- a. Ventilation system should include two sets ionized aluminum bar cooling fans each bar consists of three fans, based on transformer type and power, fixed to the base of the transformer.
- b. To achieve its proper performance, each set of fans shall be assembled together with a control and command switchboard.
- c. The cooling fans should provide optimal solutions for air forced cooling of dry transformers. The on-board ventilation system should maintain the transformer at its optimal temperature to increase the safety even in case of momentary overload. The structure of the fans should provide a fast and easy assembly / installation of the fans on the transformer enclosure.
- d. The cooling fans should have high performance crossflow and centrifugal to dissipate heat generated by transformer losses while complying and assure fail-safe operations in all zones.
- e. The cooling fans should have the following specifications:

Operation	: indoor
Degree of protection	: IP $\geq$ 20
Speed	: $\geq$ 1800 rpm
Power	: $\leq$ 250 watt (for each fan)
Noise	: $\leq$ 71 dB
Installation	: vertical / horizontal
Ambient temperature	: -5 °C to 45 °C / +50 °C (optional)
Motor insulation class	: F

5.5 Junction box for PT100 sensors

- a. The cast resin transformer shall be equipped with a box designed and manufactured for quick, safe and economic connection of PT100 sensors. The box should detect the temperature of the cast

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resin transformers.

- b. The box shall be provided with three PT100 sensors at least with not less than 2.5 m cable for each sensor. The cable length as well as the sensor type shall be determined according to the ... EDC requirements.

- c. The box should have the following specifications:

Material	: Polyamide with cable gland features
Dimensions	: 10:14 mm
Degree of protection	: IP $\geq$ 54
Flame resistance	: UL 94V0 Operating
Operating temperature	: from -5°C to +45°C (+50°C is optional)

5.6 Data cables

- a. The data cables shall be for the transport of the data signals with all the protection requirements: mod. CTES technical specifications of the extension cable for PT100.
- b. The data cables should provide the following specifications:
- 1 Cable 20 x AWG 20/19 Cu/Sn
 - 2 Flame retardant insulation PVC105
 - 3 CEI 20.35 IEC 332.1 regulations
 - 4 Maximum operating temperature: 90°C
 - 5 Conformation: 4 sets of three twisted and colored conductors
 - 6 Shield in Cu/Sn
 - 7 Flame retardant PVC sheath

5.7 Protection for temperature controller & fans protection

- a. This device should provide additional protection to protection units. The device should provide protection for the fans against over current and under voltage.
- b. This device should provide overvoltage protection and shall be used on systems where there are large surges and fluctuations in 230 Vac power supply.
- c. It shall be used for cutting the peak voltage generated when the MV switch is closed.
- d. It shall be easily installed inside the panel through the DIN EN50022 rail.
- e. It should have a red led to indicate the possible intervention of fuses.

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f. It should have the following specifications:

Power supply	: 230 Vac.
Input voltage	: 230 Vac $\pm$ 10% 50Hz
Delay time	: settable from 1 to 5 seconds
Housing material	: Polyamide UL 94V2
Protection fuses	: 2 & 5 Amps (delayed)
Ambient operating temperature	: -5 °C to +45 °C (+50 °C is optional)

6. TRANSFORMER LOSSES AND EFFECINCEY

6.1 LOSSES

The transformer shall be designed for maximum efficiency and minimum losses, when comparing between different tenders the present worth (P.W.) value of the capitalized cost of losses in the transformers shall be added to the transformer price according to the following equation:

$$P. w = K \times 8760 \times C [W_i + (L.S.F) P^2 W_{load}] L.E$$

Where,

$$K = \frac{(1+i)^N - 1}{i(1+i)^N}$$

K = Present worth factor (L.E) based on Central Bank of Egypt latest rate of interest.

i: latest interest rate from Central Bank of Egypt.

N: number of years (20 years).

C : Cost of KWH.

(According to the max commercial cost of KWH and can be adjusted as EEHC update)

LS.F = Loss factor for the load = 0.4.

W_i = Iron losses in KW at rated voltage and main tapping

W_{load} = Load losses in KW at full load and 75°C.

P = Peak load (P.U.) = 0.8.

The values of the required losses are attached in table (5)

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Table (5)

10.5 – 11 / 0.4 KV				
POWER (KVA)	IRON LOSSES (W)	LOAD LOSSES (W) at 75° C	Noise Level Lpa (db)	% Z ±10% at 75° C
50	380	1200	45	4
63	460	1400	48	4
100	590	1700	50	4
160	610	2300	50	4
250	820	3100	53	4
315	950	3600	54	4
400	1150	4300	55	4
500	1300	5100	56	4
630	1500	6400	57	4
800	1700	7700	58	5
1000	1850	8000	59	5
1250	2500	10500	61	5
1500	2600	11200	61	6
1600	2800	12300	61	6
2000	3500	14900	62	6
2500	4300	18300	65	6
3150	5500	22000	65	7
5000	8500	31500	65	7

6.2 Penalty and Rejection

a. Rejection of material:

Failure of any material to comply with any one of the requirements of this specification shall constitute grounds for rejecting that material, except if the ___EDC inspection engineer convinced that the material could satisfactorily be repaired. In such case, the contractor shall be obliged to

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replace the defective material without being entitled to any extra payment or to any extension of time to complete the contract.

b. Penalty and rejection for decreased efficiency:

1. The contractor should keep the no-load losses and load-losses to be within the guaranteed values.
2. Should the measured losses exceed the allowable tolerances, then ___EDC representative should reject the transformer with such losses. However, if the losses deviation lies within the allowable tolerances (10 % of the total losses), then the excess losses shall be evaluated as per the following formula and the contractor shall be penalized accordingly:

$$PW = K \times 8760 \times C [dW_i + (LSF) (P^2) dW_{cu}] \quad LE$$

Where

K: Present worth factor (L.E)

$$K = ((1+i)^N - 1) / i (1+i)^N$$

i: latest interest rate from Central Bank of Egypt.

N: number of years. (20 years)

C: Cost of kWh. (According to the max commercial cost of kWh and can be adjusted as per EEHC latest update)

LSF = Loss factor for the load = 0.4.

dW_i = Excess in iron (no-load) losses in kW at rated voltage and main tapping

W_{cu} = Excess in copper (load) losses in kW at full load and 95 °C.

P = Peak load (P.U.) = 0.8.

3. No credit shall be given for any decrease in the losses below the guaranteed values.

6.3 Replacement of rejected material and equipment

- In case of any rejected material or equipment, the contractor shall be obliged to replace it without extra payment.

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7. MARKING

- Transformer serial No., and rated power shall be engraved on transformer body at LV side.
- Each transformer shall be fitted with a nameplate of weather proof materials fitted in a visible position, showing the information listed below. Entries on the plate shall be indelibly marked by etching or engraving:

1. Type of transformer	2. IEC codes	3. Manufacturer's name
4. Serial number	5. Year of manufacture	6. Insulation level
7. Number of phases	8. Rated power	9. Rated frequency
10. Rated voltage	11. Rated current	12. Vector group
13. Tap position	14. Impedance voltage (% Z)	15. Type of cooling
16. Total mass	17. Temperature rise of windings	18. Customer: ... EDC
19. Short Circuit Duration	20. P.O No. / Year	21. L.V Coil Type

- If the enclosure is requested, a second similar nameplate shall be fixed by rivets on a visible position of the outside surface of the enclosure.

8. TESTING AND INSPECTION

8.1 General

- Tests shall be carried out on finished transformers in accordance with the latest relevant standard and as specified herein.
- All tests shall be carried out by and at the expense of the tenderer who shall supply all the apparatus, instruments and equipment.
- Routine tests shall be carried out in the presence and under the control of the inspecting engineer of _ _ EDC, except in specified or approved cases where test certificates are accepted. All test results shall be reviewed and approved in writing by _ _ EDC.

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- d. The contractor should advise testing dates. The contractor shall not be entitled to any extension of the time of completion due to the failure of any test or the rejection of any part of the materials as a result of any test.
- e. Routine tests shall be carried out in the supplier's factory or at any other mutually approved place.
- f. The bidder is obligated to expressly explain any difference between the submitted bid and the offered technical specification.

8.2 Type Tests

- a. The type tests shall be made on one transformer of each type in accordance with the requirements of IEC standards for Dry-Type distribution transformers.
- b. These are including and not limited to the following:
 - 1. Lightning impulse test.
 - 2. Temperature- rise test.

8.3 Special Tests

The special tests shall be according to IEC standards for dry type transformers. These are including and not limited to the following:

- 1. Measurement of sound level test
- 2. Environment test.
- 3. Climatic test.
- 4. Fire behavior test.
- 5. Short Circuit test

- The supplier shall submit short circuit test in a approved lap by the EEHC , otherwise supplier shall comply with the following:

- a. Short circuit calculations for transformer need approval shall be submit to Extra High Voltage Research Center for review with same sample transformer with its drawing.
- b. Supplier shall be guarantee transformer for 3 years, guarantee will continues to be vailed for this period or till submit short circuit test whichever is earlier

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- c. Supplier will get approval for one year only and renewed yearly if there is no complain form any --EDC during guarantee period (3 years)

8.4 Routine Tests

The routine tests shall be applied on each transformer in accordance with the requirements of IEC 60076-11 standards for dry type transformer.

These are including and not limited to the following:

1. Winding resistance test.
2. Turns ratio test and vector group.
3. Impedance voltage and load-loss test.
4. No load loss and no-load current test.
5. Applied voltage dielectric test.
6. Induced A.C withstand voltage test.
7. Partial discharge measurement (the value shall be less than 10 pc).

9. SUBMITTALS

9.1 Submittals required with tender:

- a. The tenderer shall be complete and return one legible copy of the attached Data Schedule for each rate of the offered transformers.
- b. Dimensional drawings of the offered transformers including interconnection diagrams of all components and assemblies.
- c. Original catalogues shall be submitted to facilitate evaluation of the offer.
- d. All information shall be in English language, clearly readable, with IEC standard symbols and SI units.
- e. Type and special test reports/certificates from an independent testing laboratory shall be submitted to __ EDC.
- f. Approval certificate issued by Egyptian Electricity Holding Company.
- g. List of deviations & clauses to which exception is taken (if any).

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9.2 Submittals required following award of contract:

- a. The contractor should provide with shipment of the transformer complete installation, operation and maintenance instruction/manuals suitable for the specific supplied item(s), including information from the manufacturer of each purchased accessory.
- b. It should include interconnected diagrams, drawings and instructions describing procedures for items that will be shipped to be assembled and interconnected for operation when installed.
- c. These should contain strict instructions as required for carrying, moving by skidding or rolling, and lifting the transformer and accessories.
- d. The manual should include the instructions for tap changing, enclosure assembly, disassembly, temperature indicator reading, setting procedures and other information designed to assist in proper operation and periodic maintenance of the supplied equipment.
- e. Key interlocking shall be identified, and serial number shall be supplied.
- f. All manuals, instructions and supporting drawings and parts lists supplied by the manufacturer and supplemental instruction and data supplied by accessory manufacturers shall be in English, clearly readable and arranged in such format to make the subject items easily identifiable.
- g. The contractor should provide a complete set of “as manufactured” drawings and wiring diagrams including interconnection diagrams of all components and assemblies of the equipment which assist in operation and maintenance of the equipment.
- h. All drawings shall be compatible with current industrial standards and good industrial practice.
- i. All information shall be in English language, clearly readable, with IEC standard symbols, device model, numbers and other requirements.
- j. All Dimensions shall be in centimeters.

10. SPARE PARTS (OPTIONAL)

Tender should contain a list of recommended needed spare parts.

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11. GUARANTEE

The manufacturer should guarantee the dry distribution transformer to be free from defects in both material and industry for period of 12 months from the time of operation or 18 months after delivers to ..EDC. In case of supplying for the first time, the warranty shall be 5 years starting from delivery date.

اشتراطات هامة:

1. بالنسبة للمورد/ للمُصنِّع الأجنبي:
اولاً: يجب ان يقدم سابقة خبرة خارج بلد المنشأ في ثلاث مشاريع مختلفة لمدة ثلاث سنوات تشغيل ناجح في ثلاث دول لها شبكة مماثلة في اخر 10 سنوات.
ثانياً: يجب تجربة المحول في الشبكة لمدة عام على الأقل و مراجعة اختبارات المحول قبل وبعد التجربة قبل اعتماده (وذلك بعد استيفاء اولاً).
2. يوصى بعدم تركيب هذه النوعية من المحولات في المناطق الساحلية او المناطق الصحراوية او المناطق عالية التلوث.
3. يلزم عدم تركيب هذه النوعية من المحولات البدروم او الأماكن المنخفضة التي بها مياه جوفية عالية او عرضة لتسريب المياه.

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TECHNICAL DATA SCHEDULE PARTIAL CAST Dry TYPE TR. 10.5 – 11 kV / 0.4 kV

- Manufacturer	
- Type of transformer	
- Method of cooling	
- Standard of specifications	
- Vector group	
- Winding Techniques:	
Primary	
Secondary	
- Transformer ratio at main tapping and no load	... /kV
- Transformer ratio at main tapping and full load:	... /kV
- At unity P.F.	... /kV
- At 0.8 P.F.	... /kV
- Rated primary current at main tapping and full load	A
- Rated secondary current at main tapping and full load	A
- Current density in primary winding at full load	A/mm ²
- Current density in secondary winding at full load	A/mm ²
- Primary current at main tapping and no load	A
- Flux density in core and yoke at rated voltage at main tapping and full load	Tesla
- Rated output with self-air cooling	kVA
- Rated output with forced air cooling	kVA
- Permissible overload duration without changes of specified temperature rise	
- 10% overload starting from 50% rated output	hr
- 20% overload starting from 75% rated output	hr
- 30% overload starting from 90% rated output	hr
- 40% overload starting from 100% rated output	hr
- 50% overload starting from 100% rated output	hr
- Tap changer (On load / Off load)	 /
- Number of steps	
- Voltage corresponding to each step.	kV
- Insulation levels for line terminal	
- Impulse withstand voltage + 1/50 microseconds wave (peak)	kV
- One-minute power frequency withstand voltage for primary and secondary windings (RMS)	kV
- Max symmetrical short circuit stresses withstand at rated voltage	kA
- Induced voltage test for primary winding (RMS)	kV

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- Electrical performance and characteristic	
- Resistance per phase at 75 °C - Primary winding	Ω
- Secondary winding	Ω
- Impedance voltage at full load and main tapping 75 °C	Ω
- Equivalent resistance referred to primary winding	Ω
- Equivalent reactance referred to primary winding	Ω
- Regulation at	
- Unit PF and 4/4, 3/4, 1/2, 1/4 Full load	%
- 0.8 PF lagging 4/4, 3/4, 1/2, 1/4 Full load	%
- magnetizing primary current at main tapping	A
- Partial discharge	pC
- Losses and efficiency	
- Load losses at full load and main tapping at unity PF and 75°C	kW
- Iron losses at rated voltage and main tapping	kW
- Total losses at full load and main tapping at unity PF and 75°C	kW
- Efficiency at	
140% full load at 0.8 / unity PF	%
125% full load at 0.8 / unity PF	%
100% full load at 0.8 / unity PF	%
75% full load at 0.8 / unity PF	%
50% full load at 0.8 / unity PF	%
25% full load at 0.8 / unity PF	%
- Max temperature rise at 45°C ambient at full load of	
Winding	 °C
Iron core	 °C
Hot spot	 °C
- Insulation class	
- Cooling system	
- Number of cooling units	
- Number of air fans per cooling units fan diameter /rating / and discharge No./mm/HP/1tr/min	
- Material of core	
- Material of Enclosure	
- Thickness of Enclosure	
- Sides	mm
- Cover	mm

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- Bottom	
- Insulation material	
- Primary winding	
- Secondary winding	
- Major insulation	
- Core plates	
- Number and type of coils per phase:	
- Primary windings (AL/CU)	
- Secondary windings (AL/CU)	
- Weight of core	kg
- Weight of windings	kg
- Weight of transformer ready for shipping	kg
- Dimension of transformer (without enclosure) (L× W× H)	
- Dimension of enclosure (L× W× H)	
- The minimum clearance from winding to enclosure	
- Protection degree (IP) for:	
Cooling fans	
Enclosure	
Junction box	
- Noise level	
- Climate class	
- Environmental class	
- Fire protection class	
- Vibration dampers (Yes)	
- Mechanical and thermal strength (High)	

I/we guarantee the correctness of above data for transformers we are offering to Electricity Distribution Company.

Signature of tenderer:

Date: / /